



Please write clearly in block capitals.

Centre number

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Candidate number

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Surname

Forename(s)

Candidate signature

I declare this is my own work.

A-level FURTHER MATHEMATICS

Paper 3 Mechanics

Wednesday 14 June 2023

Afternoon

Time allowed: 2 hours

Materials

- You must have the AQA Formulae and statistical tables booklet for A-level Mathematics and A-level Further Mathematics.
- You should have a graphical or scientific calculator that meets the requirements of the specification.
- You must ensure you have the other optional Question Paper/Answer Book for which you are entered (**either** Discrete **or** Statistics). You will have 2 hours to complete **both** papers.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer each question in the space provided for that question. If you require extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do **not** write outside the box around each page or on blank pages.
- Show all necessary working; otherwise marks for method may be lost.
- Do all rough work in this book. Cross through any work that you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 50.

Advice

- Unless stated otherwise, you may quote formulae, without proof, from the booklet.
- You do not necessarily need to use all the space provided.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
TOTAL	



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7367/3M

Answer **all** questions in the spaces provided.

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- 1** State the dimensions of power.

Circle your answer.

[1 mark]

$$ML^2T^{-3}$$

$$ML^3T^{-3}$$

$$ML^3T^{-2}$$

$$ML^2T^{-2}$$

- 2** The force $(3\mathbf{i} + 4\mathbf{j})\text{ N}$ acts at the point with coordinates $(0, 2)$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed along the x -axis and the y -axis respectively.

Calculate the magnitude of the moment of this force about the origin.

Circle your answer.

[1 mark]

$$6\text{ N m}$$

$$8\text{ N m}$$

$$10\text{ N m}$$

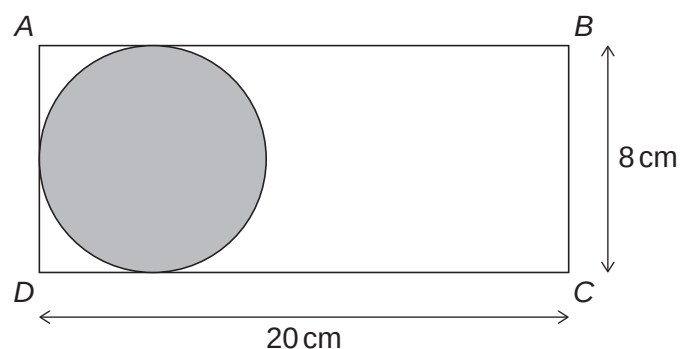
$$14\text{ N m}$$



3 A uniform disc has mass 6 kg and diameter 8 cm

A uniform rectangular lamina, $ABCD$, has mass 4 kg, width 8 cm and length 20 cm

The disc is fixed to the lamina to form a composite body as shown in the diagram below.



The sides AB , AD and CD are tangents to the disc.

Calculate the distance of the centre of mass of the composite body from AD

Circle your answer.

[1 mark]

4 cm

5.6 cm

6.4 cm

8.8 cm

4 A car of mass 1400 kg drives around a horizontal circular bend of radius 60 metres.

The car has a constant speed of 12 m s^{-1} on the bend.

Calculate the magnitude of the resultant force acting on the car.

[2 marks]

Turn over ►

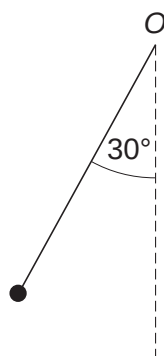


6 In this question use $g = 10 \text{ m s}^{-2}$

A sphere of mass 0.8 kg is attached to one end of a string of length 2 metres .

The other end of the string is attached to a fixed point O

The sphere is released from rest with the string taut and at an angle of 30° to the vertical, as shown in the diagram below.



6 (a) Find the speed of the sphere when it is directly below O

[3 marks]

6 (b) State one assumption that you made about the string.

[1 mark]



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- 7 (b)** State the magnitude of the impulse on B during the collision, giving a reason for your answer.

[2 marks]

- 7 (c)** Find the size of the angle between the straight line and the **impulse** acting on B , giving your answer to the nearest degree.

[2 marks]

- 7 (d)** During the collision, one particle crosses the straight line.

State which particle crosses the line, giving a reason for your answer.

[1 mark]

Turn over ►



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8 (b) State one limitation of the model that you have used.

[1 mark]

8 (c) Find the maximum speed of the block.

[4 marks]

Turn over ►



8 (d) The vertical wall is 8.7 metres from O

Determine whether the block reaches the wall, showing any calculations that you make.

[4 marks]

END OF QUESTIONS



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ANSWER IN THE SPACES PROVIDED**



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1 6

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2 3 6 A 7 3 6 7 / 3 M